SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA43

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. **(L2)**

Assessment Tool:	Code along exercise: Make your first HTML page together with CSS

Weightage: 10%

Code of Conduct:

I K. Lahari (192325118) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K. Lahari

QUESTIONS: **Total Marks: 10**

Scenario: Web Development Standards Implementation

A start up is developing a modern web application. During development, the team encounters multiple issues related to HTML elements, HTTP methods, XML syntax, and web communication protocols.

The following observations are made:

- The team needs to embed a promotional video on the homepage.
- A registration form must securely send user data to the server.
- The navigation menu must follow semantic HTML standards.
- XML configuration files are being used but contain syntax errors.
- The application must communicate over the correct web protocol.

Data Snippets Provided

Snippet 1: Video Embedding Options

A. <media src="video.mp4"></media>

B. <video src="video.mp4" controls></video>

C. <movie src="video.mp4"></movie>

D. <vid src="video.mp4"></vid>

Snippet 2: Form Submission Methods

A. GET

B. POST

C. PUT

D. DELETE

Snippet 3: Navigation Elements

A.

B. <nav>

C. <section>

D. <header>

Snippet 4: XML Syntax

A. <note><to>User</to><from>Admin</from>

B. <note><to>User</to><from>Admin</from></note>

C. <note><to>User</from></note>

D. <note to="User"></note

Snippet 5: Web Communication Protocols

A. FTP

B. HTTP

C. SMTP

D. SNMP

Questions:

1. Select the correct HTML tag for video embedding and justify your answer.
2. Identify the appropriate HTTP method for form submission and explain why.
3. Choose the proper semantic element for navigation and justify its usage.
4. Identify the correct XML syntax and explain the error in other options.
5. Choose the correct protocol for web communication and justify your choice.

QUESTION 1: Select the correct HTML tag for video embedding and justify your answer.

ANSWER 1:

Correct option: B. <video src="video.mp4" controls></video>

Explanation: The <video> tag is the standard HTML5 element for embedding video. The src attribute gives the video file path, and the controls attribute shows play, pause, volume and fullscreen buttons. It works in all modern browsers without any plugin, so it is the right choice for the promotional video on the homepage.

Why the other options are incorrect:

Option A (<media>): There is no <media> element in HTML. Browsers ignore it and will not play the video.

Option C (<movie>): <movie> is not a valid HTML tag, so the video will not be displayed.

Option D (<vid>): <vid> is not defined in any HTML standard and is not recognised by browsers.

QUESTION 2: Identify the appropriate HTTP method for form submission and explain why.

ANSWER 2:

Correct option: B. POST

Explanation: POST is the correct method for a registration form. It sends the form data in the body of the HTTP request, so the data is not visible in the URL, browser history or server logs. It also has no practical size limit and is used to create new records on the server (here, a new user account). This makes it the suitable method for securely sending user data such as names, emails and passwords (together with HTTPS).

Why the other options are incorrect:

Option A (GET): GET puts the data in the URL as a query string, so sensitive details like passwords become visible and can be stored in history and logs. It also has length limits. It is meant for retrieving data, not submitting it.

Option C (PUT): PUT is used to update or replace an existing resource. It is not used by standard HTML forms (forms support only GET and POST) and is not meant for registering a new user.

Option D (DELETE): DELETE is used to remove a resource from the server. It is not used for submitting a form and is not supported by HTML form method.

QUESTION 3: Choose the proper semantic element for navigation and justify its usage.

ANSWER 3:

Correct option: B. <nav>

Explanation: The <nav> element is the semantic HTML5 element designed to hold the main navigation links of a page, such as the menu bar. It clearly tells browsers, search engines and screen readers that the enclosed links are for site navigation, which improves accessibility, SEO and code readability.

Why the other options are incorrect:

Option A ( ): is a non-semantic generic container. It gives no meaning about its content, so it does not follow semantic standards.

Option C (<section>): <section> groups related content into a thematic section of a page. It is not specifically meant for navigation menus.

Option D (<header>): <header> holds introductory content such as a logo or title of a page or section. It can contain a <nav>, but it is not the navigation element itself.

QUESTION 4: Identify the correct XML syntax and explain the error in other options.

ANSWER 4:

Correct option: B. <note><to>User</to><from>Admin</from></note>

Explanation: Option B is well-formed XML. Every opening tag has a matching closing tag, the tags are nested properly (<to> and <from> are closed before </note>), and there is a single root element <note>. XML requires all of these rules to be followed strictly.

Errors in the other options:

Option A: The root element <note> is never closed. There is no </note> tag, so the document is not well-formed.

Option C: The <to> tag is closed with </from>, which is a mismatched closing tag. Tags must be closed with the same name and in correct order.

Option D: The closing tag is incomplete (</note is missing the ">"). Also, the attribute-based form is incorrectly terminated, so the XML is invalid.

QUESTION 5: Choose the correct protocol for web communication and justify your choice.

ANSWER 5:

Correct option: B. HTTP

Explanation: HTTP (HyperText Transfer Protocol) is the standard protocol used for communication between a web browser (client) and a web server. It transfers web pages, forms, images and videos, and uses methods such as GET and POST. Its secure version, HTTPS, adds encryption and is recommended for web applications.

Why the other options are incorrect:

Option A (FTP): File Transfer Protocol is used to upload and download files between systems. It is not used to load web pages or handle web requests.

Option C (SMTP): Simple Mail Transfer Protocol is used to send emails between mail servers, not for web communication.

Option D (SNMP): Simple Network Management Protocol is used to monitor and manage network devices such as routers and switches, not for web application communication.